Leonardo Delmiro

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SUMMARY

Senior Backend Engineer with 5+ years of experience designing, building, and operating distributed backend systems in high-scale financial environments. Strong expertise in Kotlin, Java, AWS-based backend systems, event-driven architectures, APIs, and observability. Experienced in taking solutions end-to-end, from requirements refinement and architecture design to implementation, testing, deployment, and production support. Proven track record of modernizing legacy systems, improving reliability and performance, and solving complex production issues in mission-critical financial platforms.

TECHNICAL SKILLS

Languages & Frameworks: Kotlin, Java, Python, Spring Boot, Micronaut

Architecture & Backend: Distributed Systems, Microservices, Event-Driven Architecture, REST APIs, gRPC, Clean Architecture, Hexagonal Architecture

Cloud & Infrastructure: AWS (Lambda, API Gateway, S3, SQS, SNS, EventBridge, CloudWatch), Terraform, CloudFormation, Docker, CI/CD

Data & Messaging: SQL, DynamoDB, Kafka, Redis, Amazon Athena

Observability & Reliability: Datadog, Grafana, CloudWatch, Metrics, Logging, Distributed Tracing, Alerting, Incident Response

Testing & Quality: JUnit, Kotest, Rest Assured, Automated Testing, Testcontainers

AI-Assisted Development: GitHub Copilot, Claude Code, AI Agents, SDD

PROFESSIONAL EXPERIENCE

Zup Innovation (an Itaú Unibanco Group company, the largest bank in Latin America) Remote, Brazil

Senior Software Engineer November 2023 – Present

- **End-to-End Technical Ownership:** Drove requirements refinement, solution design, and implementation of multiple features across the PIX Indirect Participants product, working closely with Product and senior engineers to evaluate alternatives, trade-offs, edge cases, and effort estimates before production delivery.
- **High-Scale Financial Systems:** Designed and developed backend services supporting the PIX Indirect Participants product, handling 1M+ daily requests with 99.9% availability.
- **Reliability Engineering:** Investigated and resolved structural production issues involving concurrent processing, missing timeouts, and inefficient retry/backoff strategies. Used logs, distributed tracing, correlation IDs, technical and business metrics, dashboards, and alerting to identify root causes and stabilize production systems, reducing production incidents by 96.25%.
- **Event-Driven Architecture**: Designed and implemented asynchronous workflows using EventBridge, SNS, SQS, and Lambda, automating participant onboarding and eliminating manual operational bottlenecks, reducing the onboarding effort by approximately 160 engineering hours.
- **Performance Optimization**: Refactored S3 data processing to asynchronous execution, reducing P90 latency from 300ms to 100ms under production load.
- **TED Modernization**: Contributed to the modernization of Itaú's TED infrastructure from mainframe to AWS by developing and testing backend services and participating in technical solution decisions for a critical financial platform processing approximately 115K transactions and R$100M in daily transaction volume.
- **Technical Leadership & Engineering Practices:** Established engineering practices through technical POCs, documentation, code reviews, and knowledge sharing. Created an automated testing approach that became a team standard, introduced Docker-based local development and debugging, and promoted AI-assisted development practices, including SDD and AI agents.

- **Mentorship**: Mentored an engineer transitioning from QA to Software Engineering over three months, providing a structured learning roadmap, pairing, code reviews, and guidance on architecture and development, enabling a successful transition into the new role.

Zup Innovation (an Itaú Unibanco Group company, the largest bank in Latin America) Remote, Brazil
Software Engineer May 2021 – October 2023
Junior Software Engineer February 2021 – April 2021

- **Backend MVP:** Designed and implemented a Kotlin backend using gRPC and REST, deployed on AWS, automating a previously manual process and reducing execution time from 7 days to less than 24 hours.
- **Legacy Modernization:** Developed Python-based AWS Lambda services and asynchronous integrations using SNS and SQS, replacing spreadsheet-driven operational workflows with automated, real-time processes.
- **Testing Strategy**: Established an automated testing approach through a technical POC, documentation, and knowledge sharing, creating a reusable model adopted as a team standard and used as a reference for subsequent projects.
- **Product Collaboration:** Collaborated with business stakeholders to refine requirements, identify edge cases, evaluate technical alternatives, estimate effort, and translate business needs into production-ready solutions.

EDUCATION

Postgraduate in Software Architecture São Paulo, Brazil
FIAP - Faculdade de Informática e Administração Paulista June 2024 – May 2025

Extension Course in Modern Software Engineering Belo Horizonte, Brazil
UFMG - Federal University of Minas Gerais March 2020 – May 2020

Bachelor's in Software Engineering São Paulo, Brazil
UNICID - Universidade Cidade de São Paulo June 2019 – June 2023

CERTIFICATIONS

AWS Certified AI Practitioner
Amazon Web Services (AWS) July 2025

AWS Certified Developer – Associate
Amazon Web Services (AWS) April 2024